

Report 1: Project Proposal

Data-Driven Growth Opportunities for Conagra in the U.S. Meat Substitutes Category

Meat Substitute Category Analysis And Assortment Optimization

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1. Executive Summary

This project evaluates the U.S. retail meat substitute category using four years of weekly POS data enriched with product-level attributes. The objective was to understand which factors drive product performance and to identify strategic opportunities for Conagra, a major participant in the plant-based meat market. We addressed three key questions:

1. Which product attributes most strongly influence sales?
2. How sensitive is consumer demand to price changes, and what price points maximize velocity?
3. How can assortment be optimized through identifying high-performing SKUs, underperformers, and white-space opportunities?

Across analyses, one insight was consistently dominant: distribution (ACV Weighted Distribution) is the single greatest driver of sales, far exceeding price, product attributes, or brand. This finding reframes how performance in this category should be evaluated and where investment should be prioritized.

We also found that price elasticity in this category is moderate for unit sales, meaning price increases do not drastically reduce sales, while volume elasticity is higher, especially among shoppers purchasing larger pack sizes. Additionally, velocity increases steeply up to about \$4.50, after which demand drops sharply. A premium pricing tier around \$6.99–\$7.25 also shows strong performance, driven primarily by premium brands like Impossible Foods.

Finally, our clustering analysis revealed four distinct SKU performance segments. A large portion of the category consists of low-distribution, low-velocity SKUs, creating assortment inefficiencies. Conagra itself holds many SKUs in these weaker clusters while also possessing a smaller number of high-performing core items deserving further expansion. The analysis highlights where Conagra should expand, rationalize, reformulate, or innovate.

The report concludes with a cohesive set of recommendations for pricing, distribution, and assortment that can materially improve Conagra's category performance.

2. Project Description

The meat substitute category is highly dynamic, with rapid innovation cycles, shifting consumer preferences, and fluctuating retailer commitment. These dynamics make it important to understand not just the trends in consumer demand, but the underlying operational levers that drive performance, such as pricing, distribution, product format, and assortment depth.

This project uses Circana/IRI-style POS datasets, which contain weekly sales, price, volume, units sold, merchandising conditions, and ACV distribution for each UPC. We combine

these with detailed product attributes including form, flavor, packaging, meat substitute type, manufacturer, and other descriptive features.

Through a combination of machine learning, econometric modeling, and clustering, we evaluate how these factors interact to shape category sales. Each section of the analysis answers a question that is meaningful not only academically but also practically for brand managers and retailer category teams.

3. Data Processing & Cleaning

Before analysis, data underwent:

3.1 Preprocessing Steps

- Converted Time → proper datetime (Week)
- Standardized column names (snake_case)
- Converted numeric fields (sales, price, ACV) to numeric types
- Filled missing sales with 0; retained NaN for prices/ACV when appropriate
- Constructed additional features:
 - log_unit_sales, log_dollar_sales
 - year, month, quarter, week_number

3.2 Final Dataset

- ~167,000 weekly UPC observations
- ~1,300 unique SKUs
- 2020–2024 timeline

4. Question 1: Product Attribute Impact on Sales

4.1 Model Overview

A Random Forest Regressor was trained on UPC-level aggregated data to predict $\log(\text{unit_sales})$ using:

- Product attributes (form, flavor, package, type, etc.)
- Size/weight measures
- Price per unit & price per volume
- ACV Weighted Distribution

The model achieved:

- $R^2 = 0.83$ (strong predictive performance)
- $RMSE = 1.31$

4.2 Feature Importance Results

Top Feature Importances (Random Forest)

- ACV Weighted Distribution — dominates with 95% of total importance
- Price per unit
- Price per volume
- Flavor (especially “Regular”)
- Total count / pack size
- Brand (Private Label effects)

Interpretation:

Distribution matters overwhelmingly more than product attributes or price.

Attribute effects exist but are secondary or tertiary.

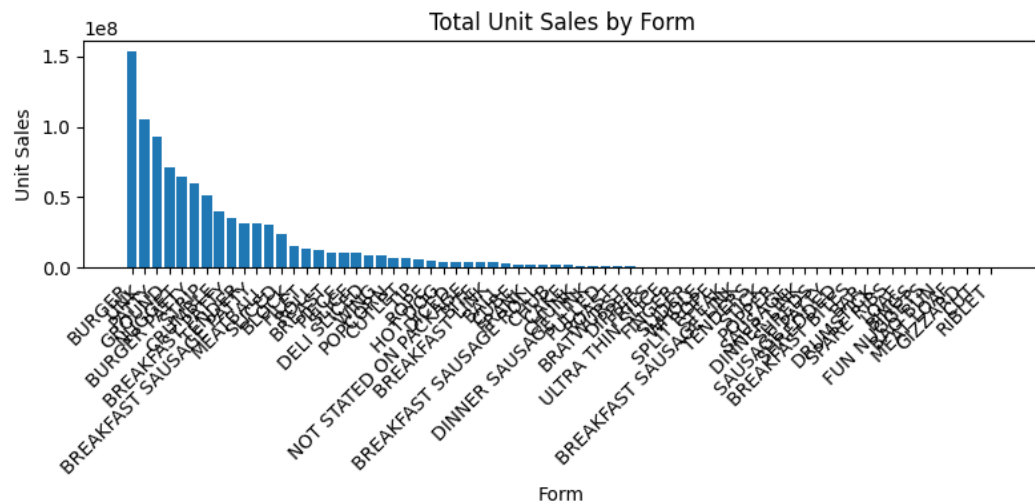




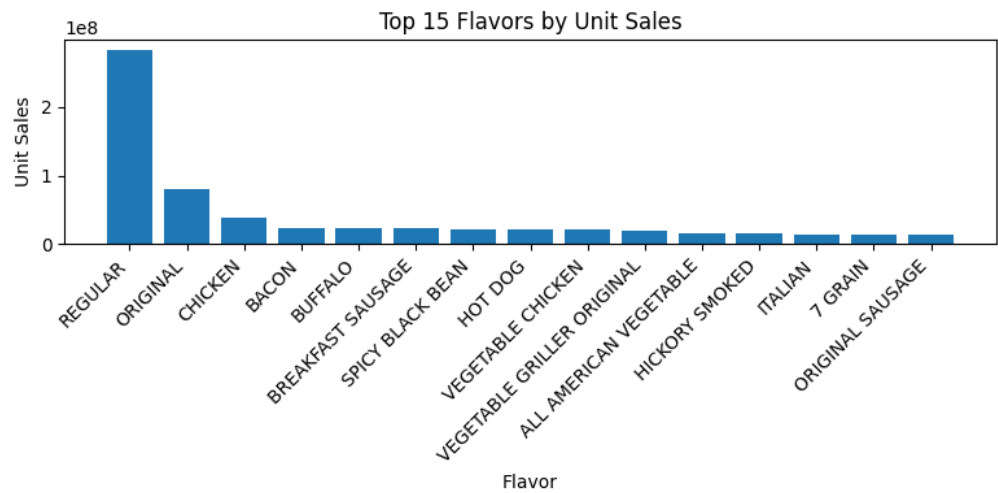
4.3 Attribute-Level Descriptive Visualizations

Total Unit Sales by Form

Burgers, ground, cutlets, tenders, meatballs dominate category demand.

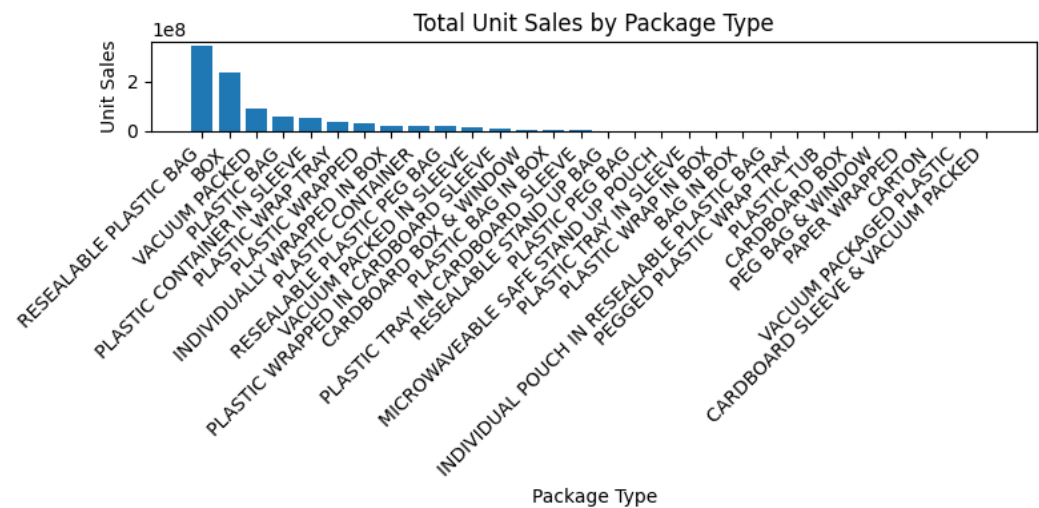


Traditional flavors (“Regular”, “Original”, “Chicken”, “Bacon”) outperform innovative flavors.



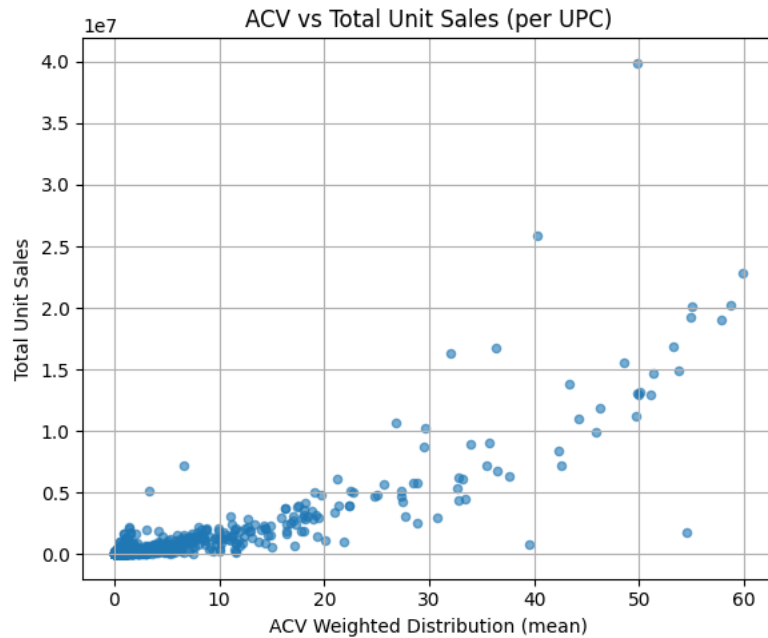
Total Unit Sales by Package Type

Resealable plastic bags are the overwhelming preferred format.



ACV vs Total Unit Sales (Scatter)

Strong linear relationship visible even without modeling.



4.4 Key Insights (Q1)

1. ACV is the #1 predictor of sales
 - a. If you're not distributed, you're not selling.
2. Price matters modestly
 - a. Higher price reduces velocity, but not dramatically.
3. Classic attributes outperform niche
 - a. Core flavors and recognizable forms dominate.
4. Packaging convenience matters
 - a. Resealable bags support repeat use and higher velocity.

5. Question 2: Pricing Elasticity & Threshold Analysis

5.1 Elasticity Model (Log-Log OLS)

Unit Sales Elasticity Results

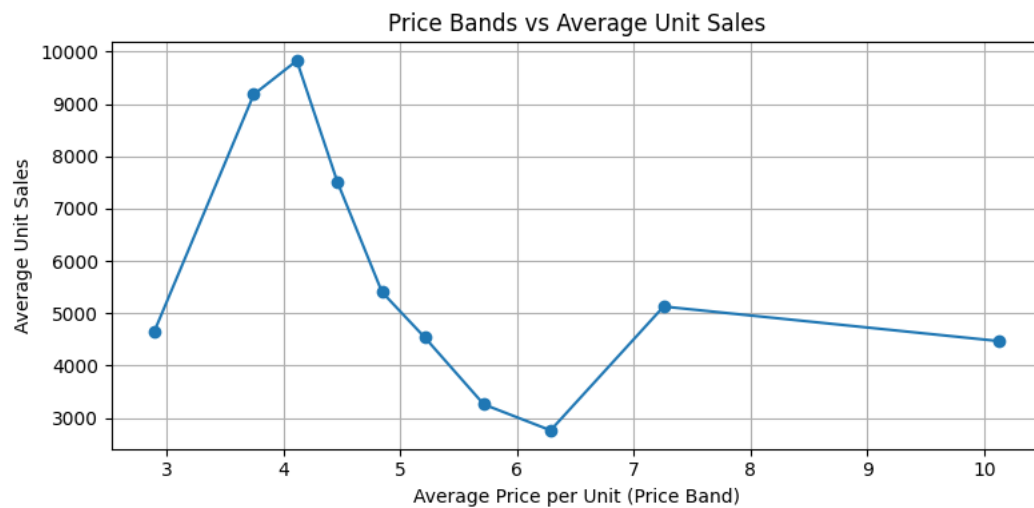
Variable	Elasticity	Interpretation
Price per Unit	-0.4435	Moderately inelastic
ACV	+0.9773	Nearly proportional lift

Volume Sales Elasticity

Variable	Elasticity	Interpretation
Price per Unit	-1.094	Elastic (value buyers are sensitive)
ACV	+0.9714	Strong driver

5.2 Price Band Analysis

Velocity peaks sharply in the \$3.50–\$4.50 price range.

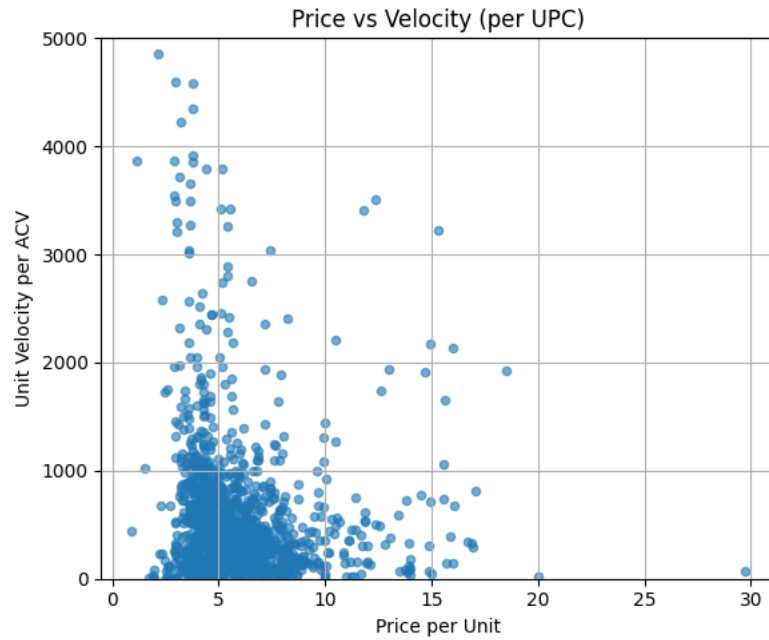


Interpretation:

- The category has a clear price cliff above \$4.75
- A premium bump exists around ~\$7 (Impossible, premium SKUs)

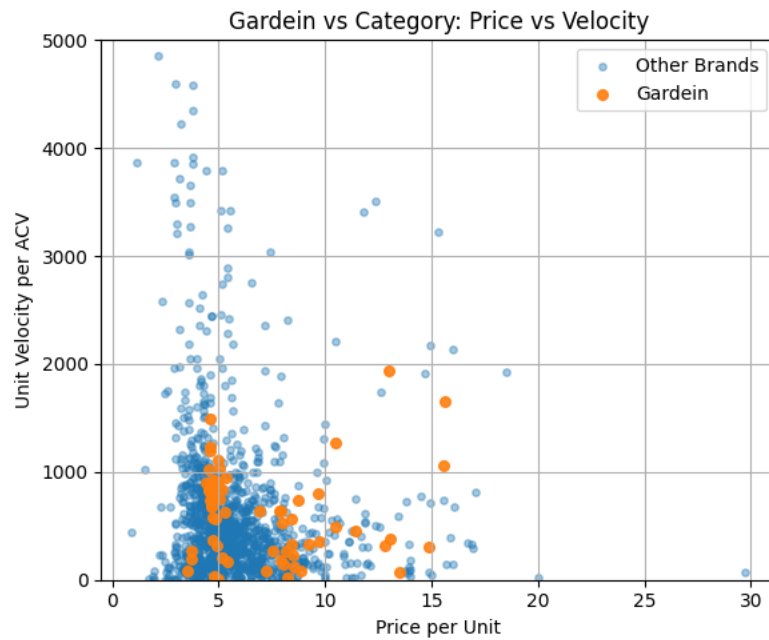
5.3 Price vs Velocity Scatterplots

Category Price–Velocity Pattern



Higher prices generally correlate with lower velocities.

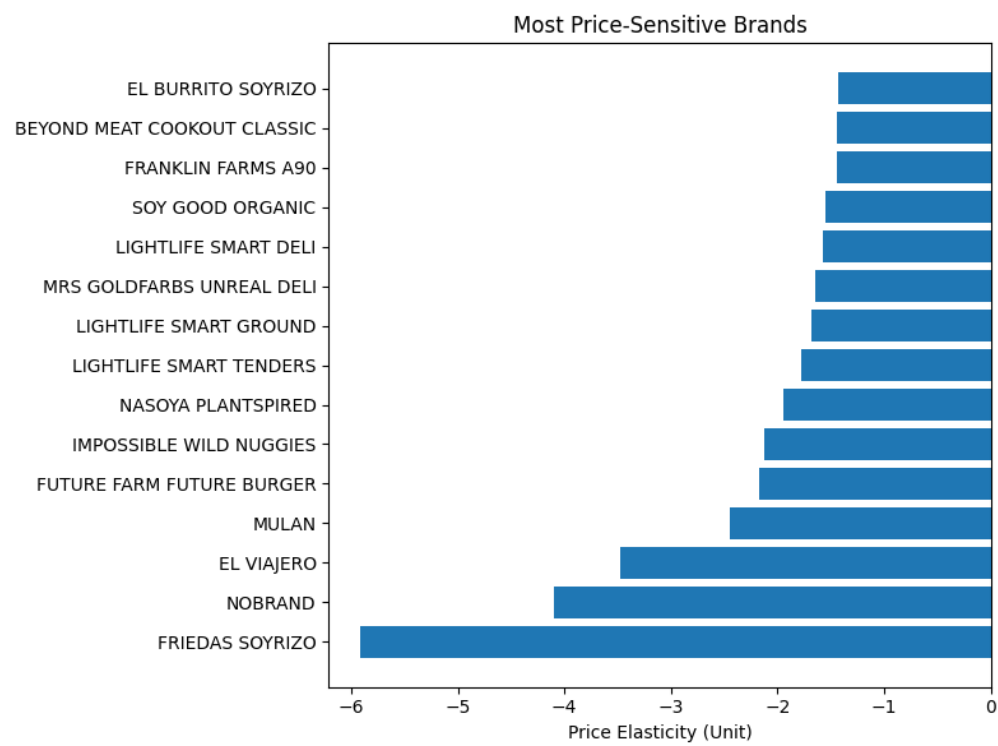
Conagra vs Category Comparison



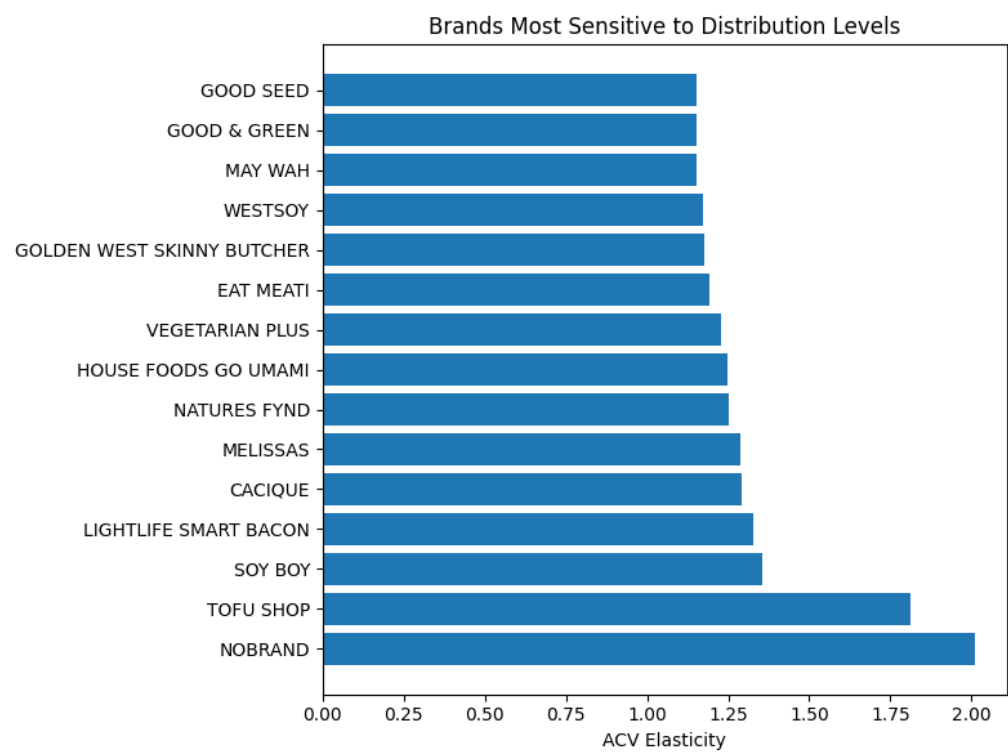
Conagra prices cluster around midrange (\$4–\$6) and align with category norms.

5.4 Brand Sensitivity Analysis

Most Price-Sensitive Brands



Most ACV-Sensitive Brands



Key takeaway:

Price sensitivity varies widely by brand, but distribution sensitivity is consistently high across the category.

5.5 Key Insights (Q2)

1. Category pricing is moderately inelastic
 - a. Price increases won't heavily damage unit sales.
2. Volume SKUs are price sensitive
 - a. Promotional pricing should focus on larger packs.
3. Optimal price band: \$3.50–\$4.50
 - a. Sweet spot for mainstream brands.
4. Distribution elasticity equals ~ 1.0
 - a. ACV remains the most powerful lever in the category.

6. Question 3: Assortment Optimization

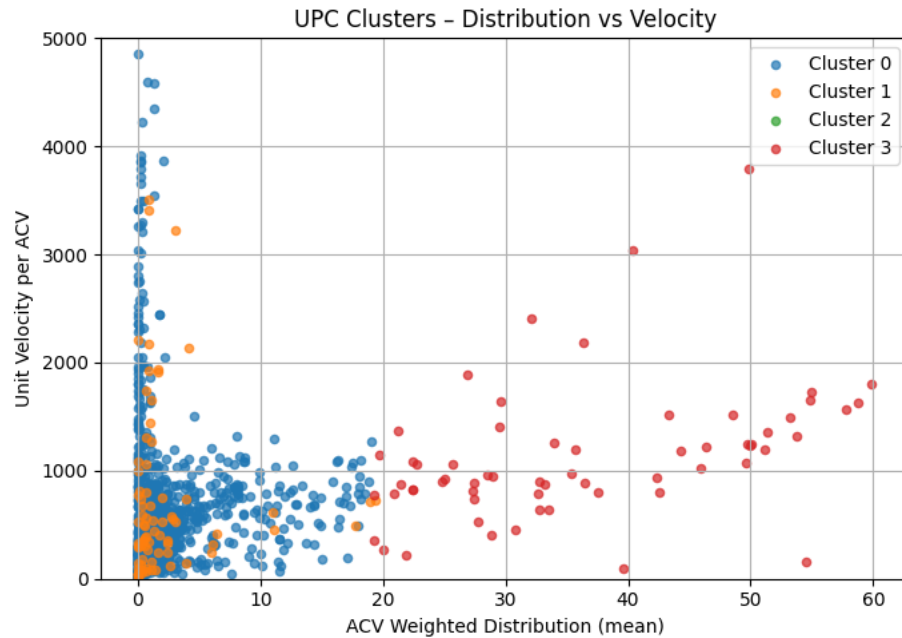
6.1 Clustering Method

K-means clustering grouped SKUs into four performance segments based on:

- Unit Velocity per ACV
- ACV Weighted Distribution
- Price per unit
- Total ounces

6.2 Cluster Visualization

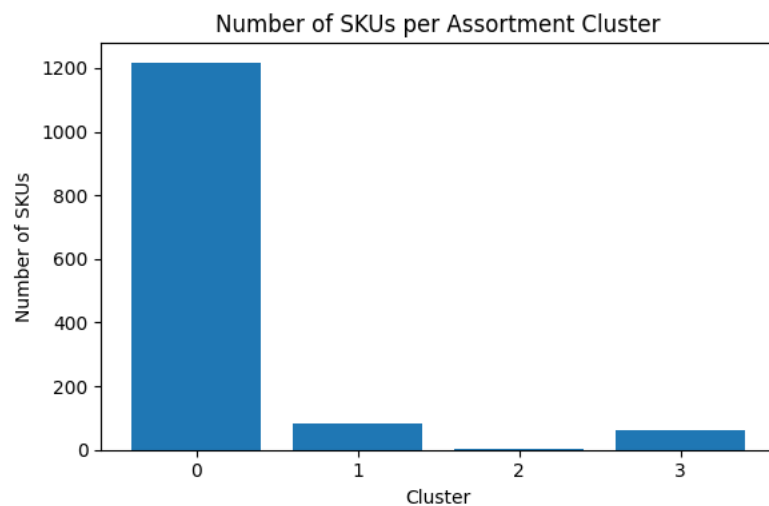
Distribution vs Velocity (Cluster Scatterplot)



Interpretation:

- Cluster 3: High-ACV, high-velocity → Core SKUs
- Cluster 2: Very high velocity, extremely low ACV → Hidden gem SKUs
- Cluster 1: Premium/niche items with high pricing
- Cluster 0: Low-velocity, low-ACV → Underperforming SKUs

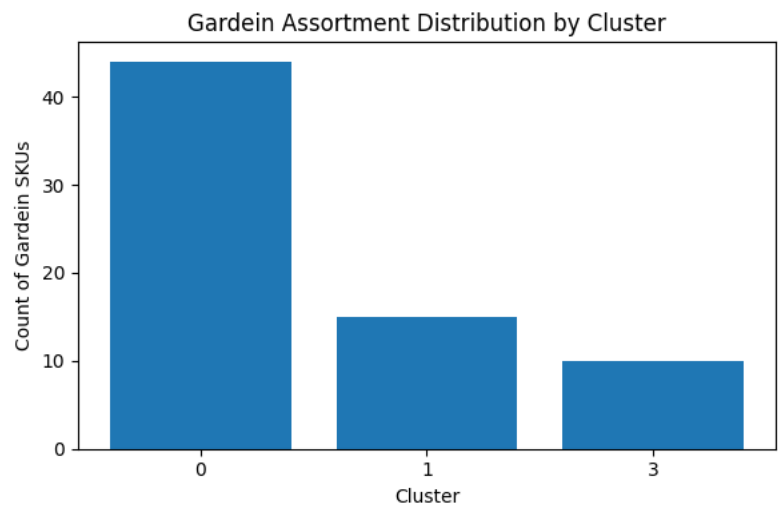
6.3 SKU Counts per Cluster



Most SKUs in the category are underperforming (Cluster 0), indicating oversaturation.

6.4 Conagra-Specific Cluster Analysis

Conagra SKUs per Cluster



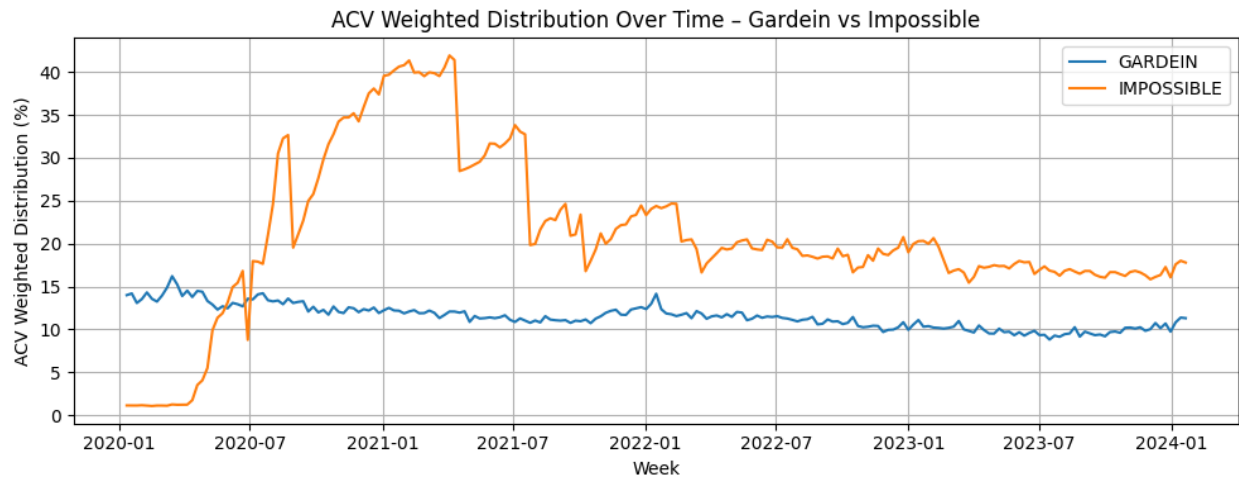
Conagra vs Category (Cluster Table)

Cluster	Other Brands	Conagra
0	1173	44
1	69	15
2	1	0
3	52	10

Interpretation:

- Conagra is over-represented in underperforming SKUs (Cluster 0)
- Conagra has no representation in Cluster 2, missing high-velocity innovations
- Conagra has 10 strong core SKUs (Cluster 3) comparable to industry leaders

6.5 Distribution Gap Analysis (Conagra vs Impossible)



Key Insight:

Impossible aggressively expanded ACV in 2020–2021, reaching >40%.

Conagra’s ACV stagnated near ~10–14%.

This distribution gap explains sales gaps more than price or attribute differences.

6.6 Key Insights (Q3)

1. Conagra is overweighted in low-performing SKUs
 - a. Rationalize Cluster 0 items.
2. Conagra has strong performers in Cluster 3
 - a. Maintain + expand ACV.
3. No presence in Cluster 2 (hidden gems)
 - a. Innovation opportunity: formats similar to category’s high-velocity winners.
4. Distribution gap vs Impossible is the #1 reason for underperformance
 - a. Fix ACV before changing pricing or attributes.

7. Recommendations

7.1 Distribution Strategy

7.1.1 Expand ACV for Core SKUs (Cluster 3)

Our analyses showed that ACV Weighted Distribution is the single strongest driver of sales, accounting for 95% of model importance in Q1 and demonstrating near-perfect elasticity (+1.0) in Q2. This means Conagra should prioritize increasing ACV for its strongest products, not necessarily creating new ones first.

Core SKUs—those in Cluster 3—already have strong velocity and consumer acceptance. Increasing distribution will directly increase sales at nearly a 1:1 rate.

Every 10% gain in ACV can be expected to yield ~10% growth in unit sales.

7.1.2 Address the Distribution Gap vs Impossible

Impossible's ACV ramped dramatically over the past four years, while Conagra's remained relatively flat. This distribution gap, not price or flavor, is the primary reason Impossible has gained category share.

To compete effectively:

- Prioritize conversations with retailers where Impossible over-indexes
- Present velocity-per-ACV comparisons to demonstrate productivity
- Secure additional facings and incremental shelf presence for top performers

7.1.3 Improve Shelf Efficiency

Retailers will support greater distribution if Conagra demonstrates category productivity. Use velocity-per-ACV and cluster classification to show where Conagra contributes incremental growth rather than redundancy.

7.2 Pricing Strategy

7.2.1 Target the Optimal Price Range of \$3.50–\$4.50

Price band analysis identified a clear velocity sweet spot. Products priced within this range experience significantly higher weekly unit sales than those priced above \$4.75.

Core recommendation:

- Keep mainstream Conagra SKUs priced between \$3.49 and \$4.49
- Avoid drifting too high, where velocity drops off

7.2.2 Limit Discounting on Core SKUs

Elasticity estimates show that unit sales are moderately inelastic (–0.44). This implies:

- Price cuts do not yield proportionally higher unit sales
- Frequent promotions erode margin without strong volume growth

Instead, Conagra should invest promotional dollars more strategically.

7.2.3 Promote Value-Size and Family Packs

Volume elasticity (-1.09) indicates that shoppers buying larger packs are more price sensitive. This means:

- Promotions will be more effective on multi-serve SKUs
- Incremental units from price drops are more substantial in this tier
- Retailers often support value-pack promotions due to larger basket rings

7.2.4 Consider a Premium Tier at \$6.99+

The price threshold analysis revealed a premium bump around \$7—a zone dominated by Impossible.

This segment reflects:

- Consumers willing to pay for engineered, high-quality plant proteins
- Strong brand loyalty
- Higher margins

Conagra can enter this space with:

- More culinary-oriented SKUs
- Higher protein content
- New formulations or “chef-crafted” concepts
- Packaging that signals a premium experience

This allows Conagra to broaden its appeal without eroding mainstream equity.

7.3 Assortment Strategy

7.3.1 Rationalize Underperforming SKUs (Cluster 0)

Conagra has 44 SKUs in Cluster 0, meaning they show:

- Low velocity
- Low ACV
- Limited consumer traction

Retailers often push for category simplification. Removing or reformulating these SKUs will:

- Improve brand productivity
- Free up shelf space for stronger items
- Reduce supply chain and manufacturing complexity
- Make room for innovation with better performance potential

7.3.2 Expand Distribution for Core Winners (Cluster 3)

Conagra's 10 Cluster 3 SKUs have strong velocity and broad appeal. These should be:

- Expanded into more stores
- Increased in facings
- Given stronger promotional support
- Highlighted in retailer line reviews

These SKUs form the foundation of Conagra's brand strength.

7.3.3 Innovate Into Missing White Space (Cluster 2)

Cluster 2 is the most interesting insight from Q3:

- These SKUs exhibit very high velocity,
- But extremely low ACV,
- Meaning they perform exceptionally well where placed.

Conagra currently has no SKUs in Cluster 2, representing a major whitespace opportunity. Products in this cluster tend to have:

- Versatile formats (e.g., ground, strips)
- Broad flavor appeal
- Strong price–value alignment

We recommend Conagra explore launching new items modeled on these characteristics.

7.3.4 Strengthen Premium and Niche Offerings (Cluster 1)

Conagra's 15 SKUs in Cluster 1 show moderate velocity at higher price points. These items serve:

- Premium shoppers
- Specialty cooking needs

- Expanded culinary occasions

Conagra should maintain a curated set of premium offerings but avoid overexpanding this tier unless supported by strong demand.

7.4 Innovation Strategy

7.4.1 Prioritize High-Performing Formats

These include:

- Ground products
- Strips and tenders
- Classic burger and cutlet formats

These formats appear consistently across high-performing clusters.

7.4.2 Lean Into Staple Flavors

Classic, familiar flavors dominate the category:

- Original
- Chicken-style
- Beef-style
- BBQ
- Teriyaki

New innovations should stay within these guardrails for broadest market acceptance.

7.4.3 Use Resealable Packaging

Consumer preference is clearly skewed toward convenience, and this package type correlates with higher velocities.

7.4.4 Consider Functional Claims

To enter the premium space, innovation may integrate:

- Higher protein
- Cleaner ingredient decks
- Cooking versatility

- Restaurant-inspired flavor systems

8. Conclusion

This project provides a comprehensive evaluation of the plant-based meat category using advanced analytical methods. Through machine learning, econometric modeling, and performance clustering, we gained deep insight into the forces shaping retail outcomes in this fast-evolving space.

Across all results, distribution emerges as the single most important determinant of success, dwarfing the impact of product attributes and even pricing. This challenges the common assumption that innovation alone drives growth. Instead, availability—measured through ACV Weighted Distribution—plays an outsized role. To grow meaningfully, Conagra must focus on closing its distribution gap with competitors like Impossible Foods.

Pricing analysis reveals a category with moderate price sensitivity and clear thresholds. Maintaining prices within the \$3.50–\$4.50 range maximizes velocity, while selectively pursuing a premium tier offers additional margin opportunities. The limited elasticity also indicates that deep discounts are not required for growth and may erode margins unnecessarily.

Assortment analysis highlights significant inefficiencies in the category. Conagra holds a large number of underperforming SKUs but also possesses strong core items worthy of expansion. The lack of presence in the high-velocity white-space cluster (Cluster 2) presents a compelling innovation opportunity that should be explored strategically.

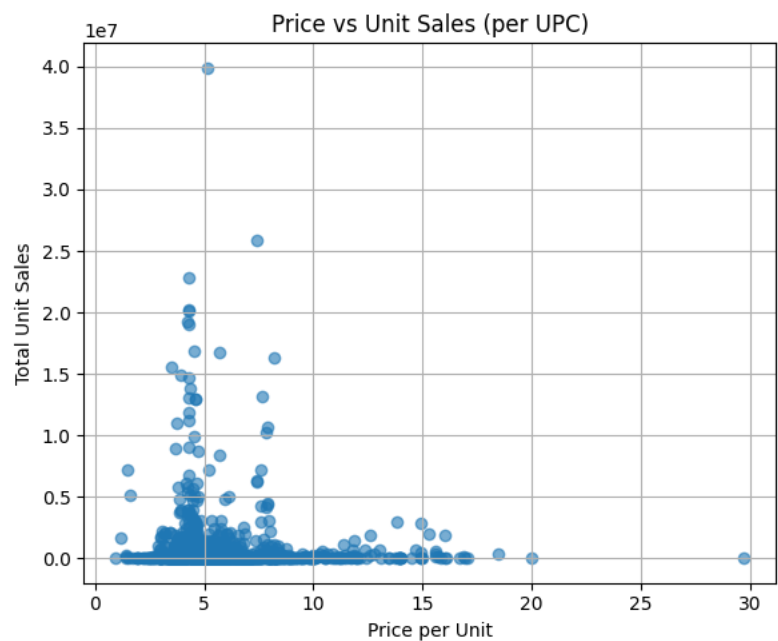
Taken together, the findings support a cohesive growth strategy:

- Drive distribution expansion for core SKUs
- Rationalize underperformers to improve productivity
- Align pricing with category sweet spots
- Innovate into proven formats and flavor spaces
- Pursue premium opportunities selectively

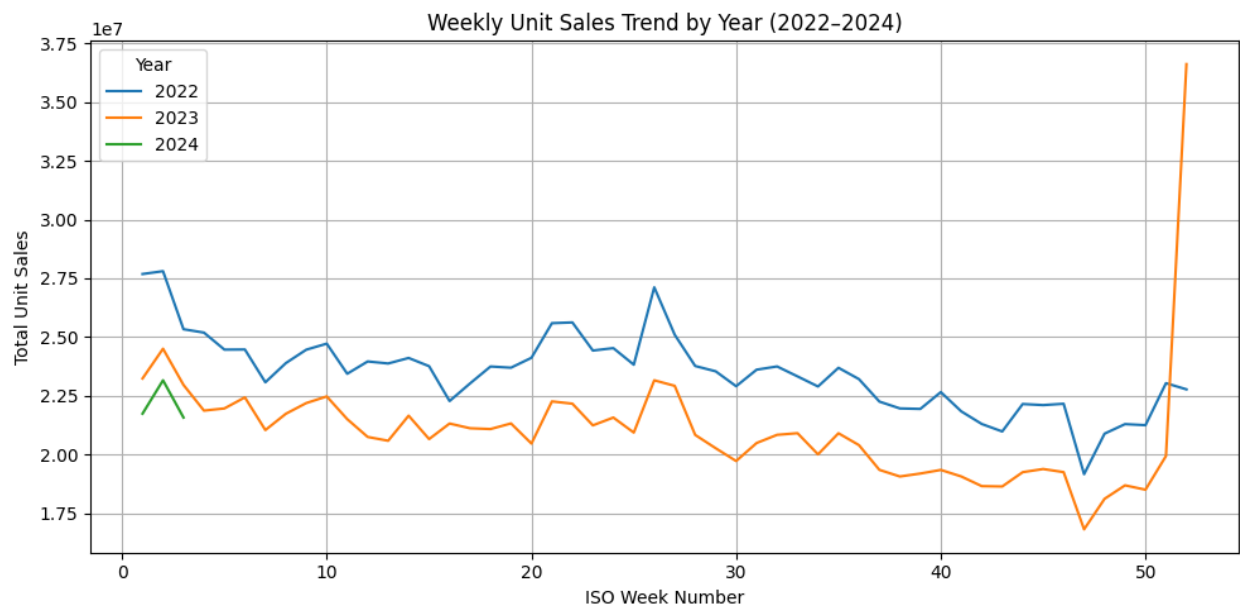
By grounding decisions in data and prioritizing the most impactful levers—distribution, pricing discipline, assortment efficiency, and targeted innovation—Conagra can strengthen its competitive positioning and achieve sustainable long-term growth in the meat substitute category.

Appendix

Graph 1: Price vs Unit Sales (per UPC)



Graph 2: Weekly Unit Sales Trend by Year (2022 – 2024)



Graph 3: Weekly Sales Trend for Product Yearly

